



Bluestock Mutual Fund Analysis

Final Capstone Report

Industry context, fund performance, risk, investor behaviour, and an interactive Streamlit dashboard.

Scope	Evidence in this report
Datasets	10 supplied CSV sources, transformed to a SQLite-backed analytical model
Fund sample	40 schemes; 40 with observed NAV
Analysis window	Daily NAV observations: 2022-01-03 to 2026-05-29
Interactive output	Four-page Streamlit dashboard; no hosted URL or native PBIX is claimed

Prepared from the project datasets and generated analysis outputs.

1. Executive summary

The project combines industry snapshots, 40-scheme NAV history, performance and risk analytics, a 32,778-row investor transaction sample, and an interactive dashboard. The latest supplied AUM snapshot (Dec 2025) totals INR 62.74 lakh crore across reported fund-house records. December 2025 SIP inflow is INR 31,002 crore; the latest folio snapshot records 26.12 crore.

Headline	Result	Interpretation
Scorecard leader	Mirae Asset Large Cap Fund - Regular - Growth	Composite score 86.25/100
Lowest 95% daily VaR	SBI Small Cap Fund - Direct Plan - Growth	-2.69% threshold
Fund coverage	40 schemes	Sample, not a census of all Indian mutual funds
NAV observation dates	1,608	Market-date series used for return analysis

The strongest practical outcome is a reproducible local analytics workflow: cleaned data, SQLite model, notebooks and scripts, downloadable static exports, and a Streamlit dashboard. Performance rankings are descriptive and should not be read as investment advice.

2. Objectives and completion review

Objective	Project evidence	Status
Data ingestion and validation	10 CSV inputs; code/NAV coverage checks; live NAV fetch utility	Complete
Cleaning and analytical model	Processed datasets, SQLite database, schema and query SQL	Complete
EDA and performance	EDA, performance and advanced analytics notebooks/reports	Complete
Risk and investor analytics	VaR/CVaR, rolling Sharpe, cohort/SIP continuity, HHI, recommender	Complete
Interactive dashboard	Four-page Streamlit app with filters and charts	Complete
Native Power BI PBIX	Generated PBIP was not validated in Desktop	Not delivered
Public dashboard publishing	No hosting/account destination configured	Not published

This review distinguishes implemented local deliverables from optional platform deliverables. The Streamlit app is the supported interactive route in this repository; neither a published URL nor a validated Power BI PBIX is represented as complete.

3. Data sources and coverage

The supplied local CSVs are the analytical source of truth. A live NAV helper also queries MFAPI for selected public scheme histories; the pipeline does not make external requests by default.

Source file	Purpose	Observed coverage / shape
01_fund_master.csv	Scheme metadata	40 rows
02_nav_history.csv	Daily scheme NAV	46,000 rows; 40 schemes
03_aum_by_fund_house.csv	AMC-level AUM snapshots	90 rows; 10 AMCs
04_monthly_sip_inflows.csv	Industry SIP trends	48 months
05_category_inflows.csv	Category flows	144 rows; FY25 coverage
06_industry_folio_count.csv	Folio trends	21 monthly snapshots
07_scheme_performance.csv	Scheme performance attributes	40 rows
08_investor_transactions.csv	Synthetic/sample transaction records	32,778 rows; Jan 2024-May 2025
09_portfolio_holdings.csv	Scheme holdings and sector weights	322 rows; 34 schemes
10_benchmark_indices.csv	Market/benchmark index levels	8,050 rows

The datasets have different reporting windows and grains; dates were not artificially aligned for all analyses. Investor data is treated as a provided sample, not a representative population survey.

4. Data quality and validation

All 40 distinct fund-master AMFI codes are present in the raw NAV history. The master contains 40 unique codes and NAV covers 40 distinct codes. AMFI scheme codes are opaque identifiers; their digits are not interpreted as decoded attributes.

Check	Observed result	Handling
AMFI code coverage	40/40 master codes found in NAV	Exact key coverage validated
Calendar-filled NAV	Cleaned NAV includes forward-filled dates	Returns/risk use observed market dates
Five-year history	Unavailable within observed 2022-2026 span	Five-year CAGR reported as NA
SIP YoY growth	12 initial 2022 values absent	Retain missing; do not impute
Category inflows	FY25 only	Limit category comparisons to available year
Holdings snapshot	One sample snapshot; 34 schemes	Sector HHI is snapshot-specific

Forward-filled calendar NAV adds non-trading dates and can create artificial zero returns. It remains useful for continuity and dashboard display, but is excluded from the daily return-based risk calculations.

5. ETL and analytical data model

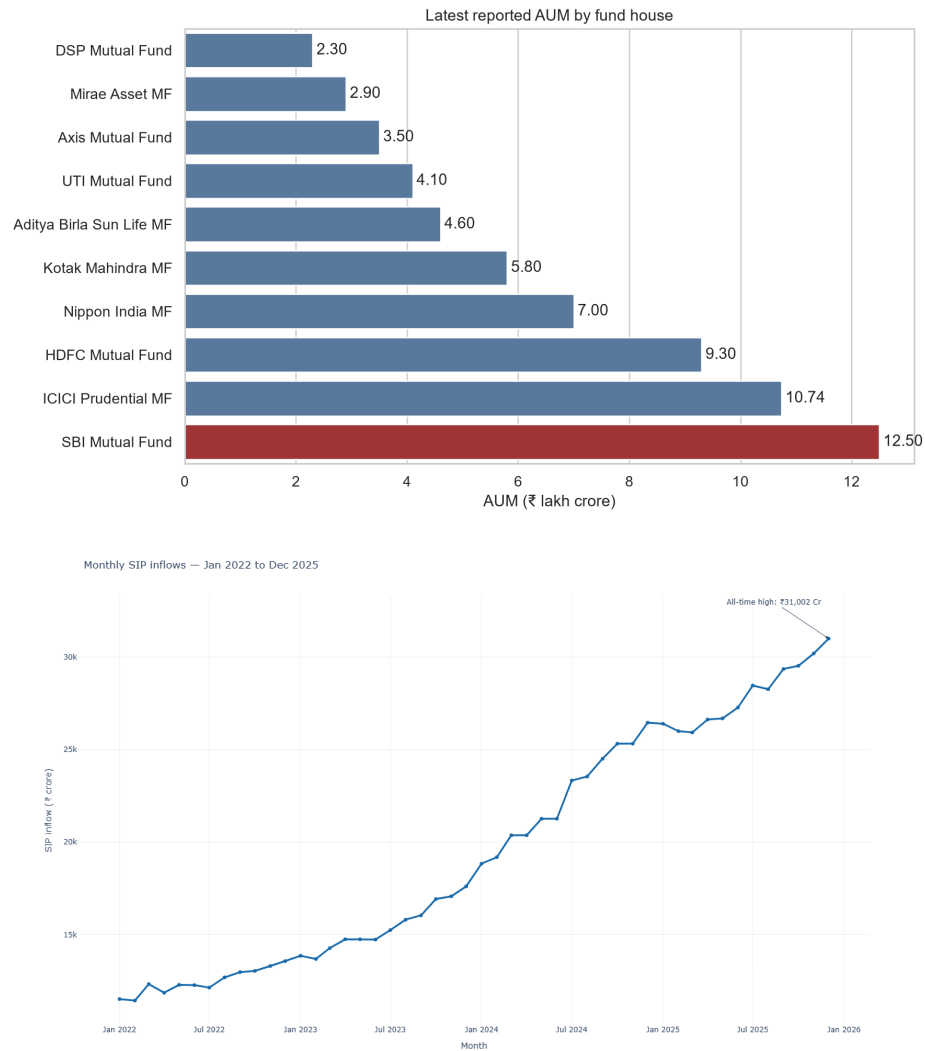
The ETL reads raw CSV files, normalizes types and identifiers, performs data-quality checks, writes processed CSVs, and loads the project SQLite database. SQL schema and query files document the resulting relational model.

Layer	Responsibilities	Key outputs
Raw	Preserve received CSVs; optional live NAV utility	data/raw/
Transform	Normalize dates/codes, clean fields, validate keys	day2_etl.py
Processed	Clean analysis-ready tables	data/processed/*.csv
Storage	Relational tables and constraints	bluestock_mf.db, schema.sql
Analytics	EDA, returns/risk, investor and holdings analysis	notebooks/, analytics/
Presentation	Interactive and static dashboards/reports	dashboard/, reports/

The `amfi\_code` field links scheme-level tables and `date` links dated observations. Transaction, NAV, benchmark and snapshot datasets retain their own natural grains; joining them requires explicit aggregation and date alignment.

6. EDA: industry scale and flows

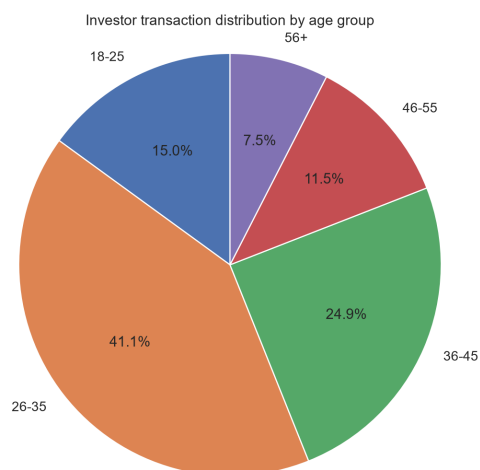
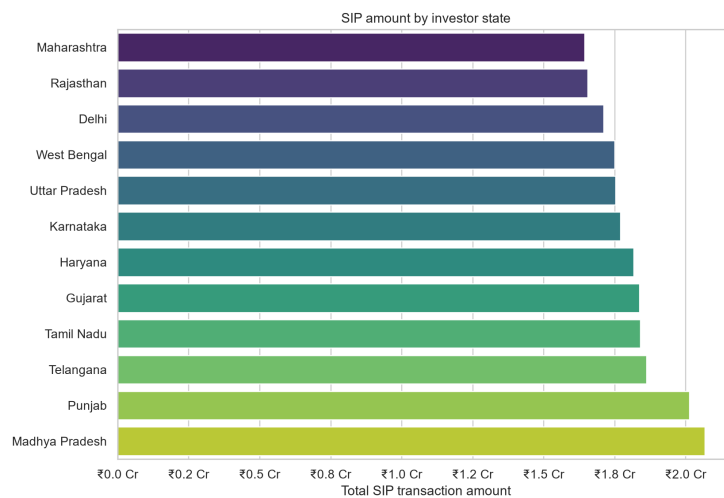
The latest AUM date in the supplied fund-house snapshots is December 2025; combined reported AUM is INR 62.74 lakh crore. The December 2025 SIP series ends at INR 31,002 crore, its maximum observation in the supplied 48-month series.



Source: supplied AUM and SIP snapshots. Amounts are reported values, not forecasts.

7. EDA: investor activity and geography

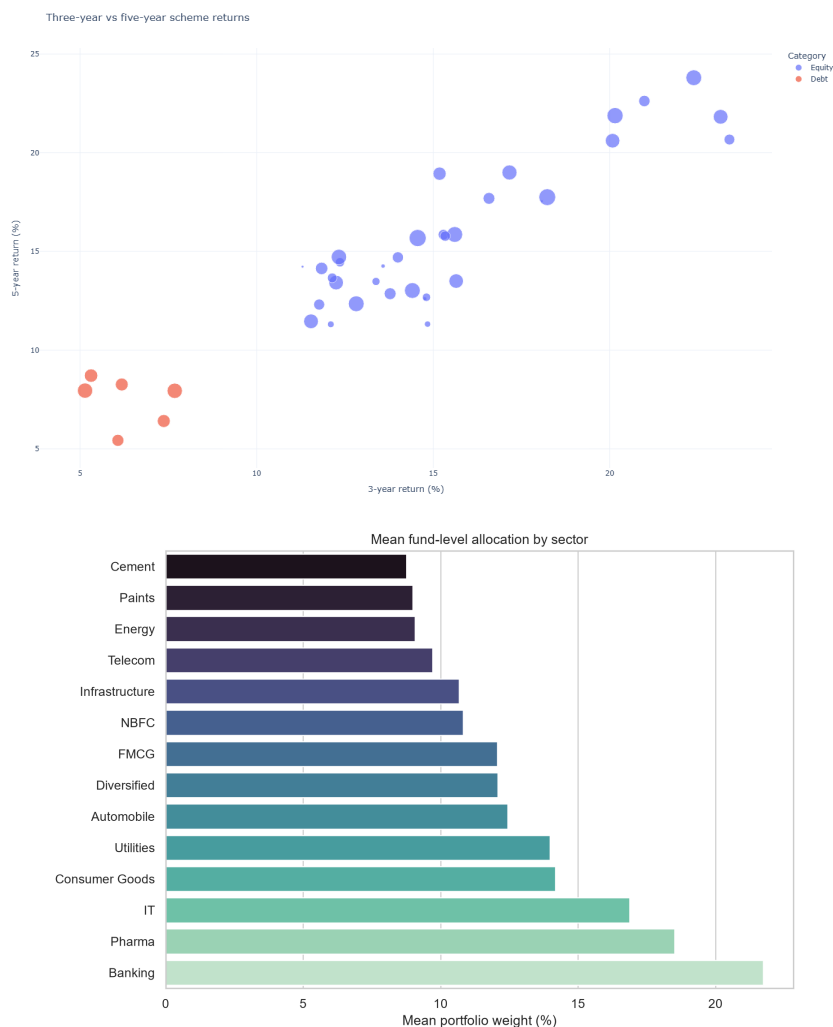
The transaction sample contains 32,778 records over Jan 2024 to May 2025. State, city-tier, age-band and transaction-type views describe these records only.



Transaction totals are sample summaries and are not estimates of state-level market share.

8. EDA: scheme and portfolio structure

Scheme performance and expense comparisons show variation across the available sample. The category-flow data spans FY25 only; holdings represent one snapshot, so these figures should not be generalized to current portfolios.



9. Performance methodology

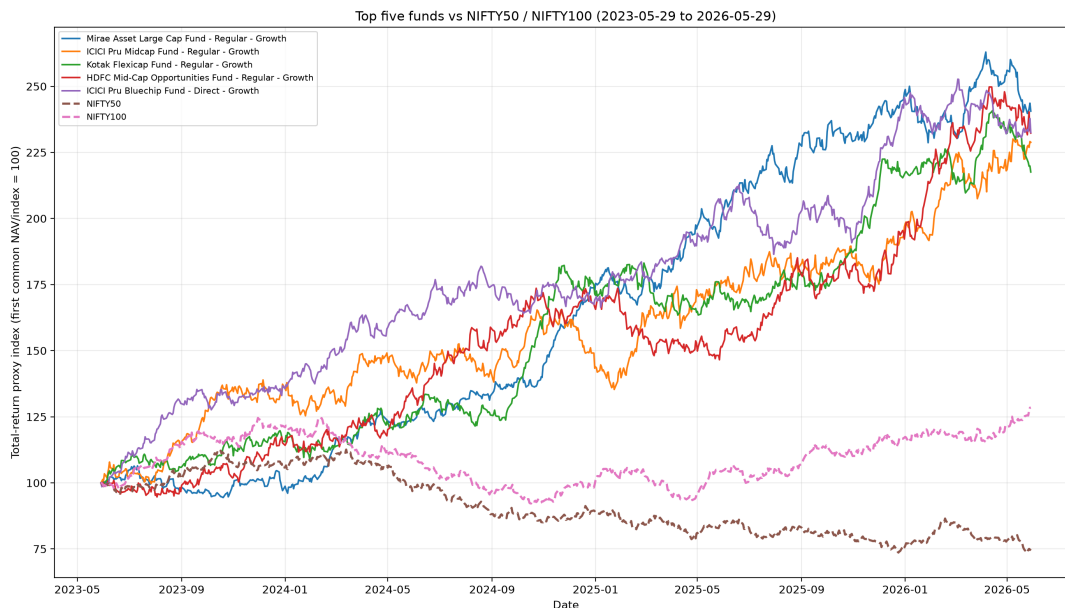
Metric	Definition / implementation
Daily return	$\text{NAV}(t) / \text{NAV}(t-1) - 1$ on common observed market dates
CAGR	$(\text{NAV_end} / \text{NAV_start})^{(1/\text{years})} - 1$; 1y/3y lookback, 5y unavailable
Sharpe	Annualized mean daily excess return / sample SD; risk-free rate 6.5% p.a.
Sortino	Annualized excess return / downside deviation of negative excess-return days
Alpha / beta	OLS on daily fund vs NIFTY100 returns; alpha annualized as intercept x 252
Max drawdown	Minimum NAV / running peak - 1; peak/trough and recovery recorded
Tracking error	Sample SD of active daily returns x $\sqrt{252}$

For the return analysis, 1,150 common market dates provide 1,149 daily observations per fund. The annualization convention follows 252 trading days. Beta, alpha and tracking error remain sample- and benchmark-dependent estimates.

10. Performance results and composite scorecard

The scorecard combines percentile ranks for three-year return (30%), Sharpe (25%), annualized alpha (20%), inverse expense ratio (15%), and inverse maximum drawdown (10%). It is a transparent ranking heuristic, not an expected-return model.

Rank	Scheme	3y CAGR	Sharpe	Alpha (ann.)	Score
1	Mirae Asset Large Cap Fund - Regular - Gro	34.0%	1.46	27.0%	86.2
2	ICICI Pru Midcap Fund - Regular - Growth	31.8%	1.19	29.3%	82.2
3	Kotak Flexicap Fund - Regular - Growth	29.6%	1.32	27.3%	82.0
4	HDFC Mid-Cap Opportunities Fund - Regular	32.4%	1.10	27.2%	80.8
5	ICICI Pru Bluechip Fund - Direct - Growth	32.5%	1.04	21.2%	80.0
6	Axis Midcap Fund - Regular - Growth	35.1%	1.01	26.1%	77.0

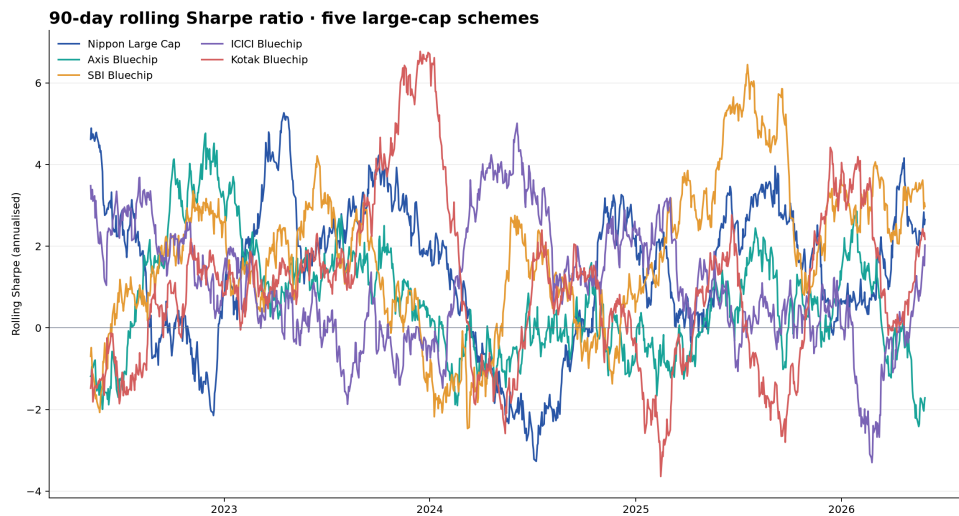


Top-five composite-score funds and NIFTY 50/NIFTY 100 are rebased to 100 over the latest available three-year common window. Past relative performance is not predictive.

11. Risk analysis: tail losses and rolling Sharpe

The most negative historical 95% one-day VaR in this sample is for SBI Small Cap Fund - Direct Plan - Growth: -2.69%; its conditional tail mean is -3.24%. Historical VaR/CVaR are distribution summaries, not loss guarantees.

Fund	Risk grade	95% VaR (daily)	CVaR (daily)
SBI Small Cap Fund - Direct Plan - Growth	Very High	-2.69%	-3.24%
Axis Small Cap Fund - Regular - Growth	Very High	-2.62%	-3.17%
ABSL Small Cap Fund - Regular - Growth	Very High	-2.60%	-3.25%
Nippon India Small Cap Fund - Regular - Grow	Very High	-2.54%	-3.23%
SBI Small Cap Fund - Regular Plan - Growth	Very High	-2.45%	-3.06%
DSP Small Cap Fund - Regular - Growth	Very High	-2.35%	-3.10%



Rolling Sharpe uses a 90-observation window and annualization by $\sqrt{252}$.

12. Investor cohorts, SIP continuity and concentration

Investor-level analyses group the sample by first transaction year and summarize investment amounts and scheme preference. SIP continuity applies the stated eligibility of at least six SIP transactions; investors with mean inter-transaction gaps above 35 days are marked at-risk.

Analysis	Result	Caveat
Cohorts	First transaction year; average SIP and total invested	Sample-period only
SIP continuity	6+ SIP transactions; mean gap >35 days = at-risk	Same-date duplicate rows can create zero-day gaps
At-risk SIP flag	1,332 of 1,362 eligible investors = 97.80%	Rule-based flag, not churn probability
SIP continuity rate	2.20%	Complement of the stated at-risk classification
Sector concentration	HHI = sum of squared sector weights per scheme	Single holdings snapshot; aggregation is sector-level
Highest observed HHI	Axis Bluechip Fund - Regular - Growth: 2967.7	Higher HHI means more concentrated by this measure

These rules support monitoring and exploration; they do not establish investor intent or causal reasons for missed contributions.

13. Dashboard: industry and fund performance

The Streamlit dashboard provides four interactive sections with fund-house, category, plan and date filters where relevant. The industry page summarizes AUM, SIP inflow, folios and reported scheme counts; the performance page provides fund comparisons.

Industry Overview

Latest available source data; industry and AMC-reported measures are not the 40-scheme sample.



₹62.74L Cr

Latest AUM - Dec 2025

₹31,002 Cr

Monthly SIP - 2025-12

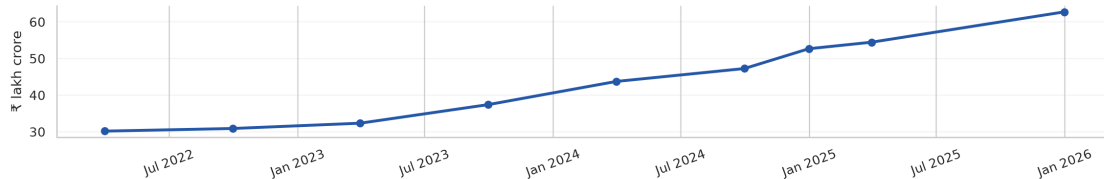
26.12 Cr

Industry folios - 2025-12

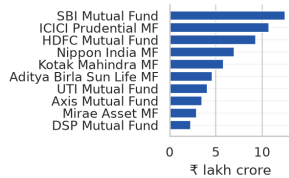
1,522

AMC-reported schemes - latest snapshot

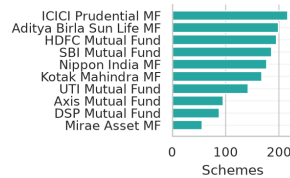
Industry AUM trend (reported snapshots)



Latest AUM by asset manager



AMC-reported scheme count



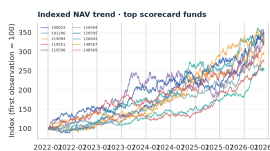
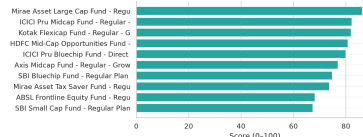
Bluestock Mutual Fund Analysis | Source: supplied project datasets | Values reflect available sample coverage

Fund Performance

Bluestock Mutual Fund Analysis | Source: supplied project datasets | Coverage starts in January 2022.



Top 10 funds by composite score



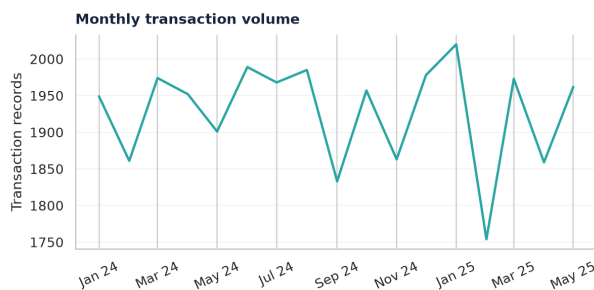
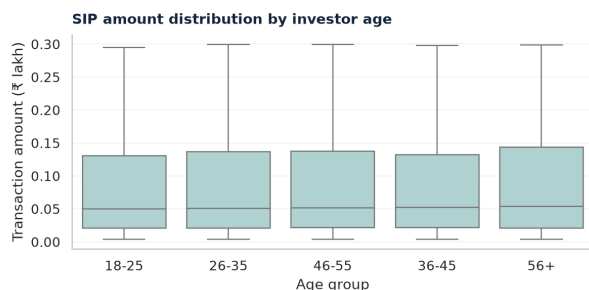
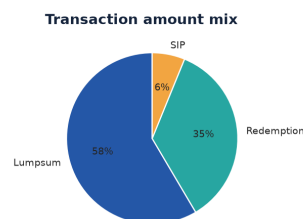
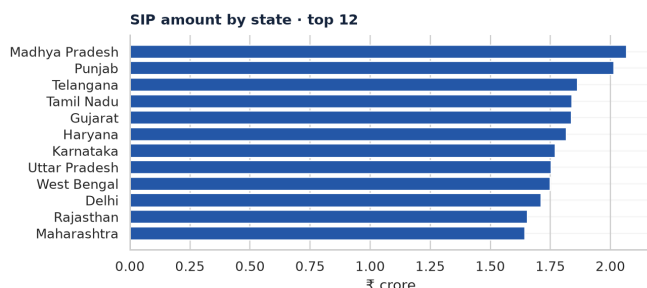
Bluestock Mutual Fund Analysis | Source: supplied project datasets | Values reflect available sample coverage

14. Dashboard: investor analytics and market trends

Investor analytics shows sample transaction geography, mix and demographic segments. The SIP/market page compares monthly SIP and market context and includes category flow views. Charts are limited by the coverage of their source tables.

Investor Analytics

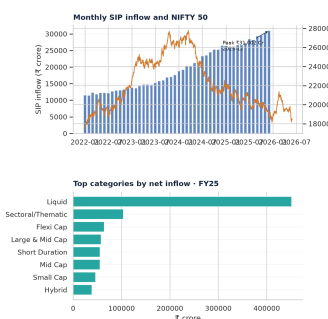
Transaction sample available through May 2025; amounts are based on supplied transaction records.



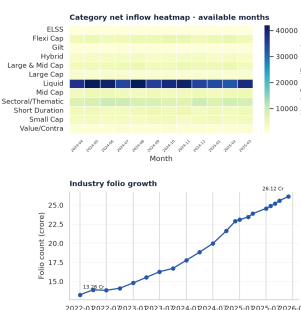
Bluestock Mutual Fund Analysis | Source: supplied project datasets | Values reflect available sample coverage

SIP & Market Trends

SIP and Lumpsum inflows through December 2023; category inflows cover FY23.



Bluestock Mutual Fund Analysis | Source: supplied project datasets | Values reflect available sample coverage



15. Limitations, recommendations and sign-off

Limitation	Recommended next step
40-scheme sample; no complete market universe	Expand scheme coverage and benchmark mapping
Five-year history unavailable	Extend observed NAV history before publishing five-year CAGR
Category inflows FY25 only; holdings snapshot only	Add recurring snapshots and explicit as-of dates
Investor sample is not population-representative	Document sampling design and protect privacy
Historical metrics and arbitrary score weights	Add sensitivity checks and suitability review before decisions
Dashboard is local; no PBIX/hosted URL	Publish only after approval, access controls and data review

Recommendation: use the scorecard and risk outputs as screening aids, then validate holdings, benchmark fit, fees, time horizon, and investor suitability. Re-run the pipeline when source datasets are updated; keep observed and imputed NAV clearly distinguished.

Self-review: local ETL, analytics, Streamlit dashboard, PDF report and presentation are included. Native Power BI PBIX and public hosting remain explicitly out of scope/unverified.